

The decoder is therefore a stochastic function:

$$z \rightarrow \text{distribution over } x.$$

The variability in outputs comes from:

- Sampling different z
- Sampling from $p(x|z)$

Why We Cannot Optimize Directly

The marginal likelihood:

$$\log p_{\theta}(x) = \log \int p_{\theta}(x \mid z) p(z) dz$$

is usually intractable.

So instead, we maximize a lower bound:

$$\text{ELBO} = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x \mid z)] - \text{KL}(q_{\phi}(z \mid x) \parallel p(z)).$$

Maximizing the ELBO indirectly pushes:

$$p_{\theta}(x) \rightarrow p_{\text{data}}(x).$$

Generative Interpretation

After training, the model generates data by:

1. Sample: $z \sim p(z)$
2. Decode: $x \sim p_{\theta}(x \mid z).$

The decoder should produce outputs x whose overall distribution satisfies:

$$p_{\theta}(x) = \int p_{\theta}(x \mid z) p(z) dz \approx p_{\text{data}}(x)$$

